Question:

A 4-year-old boy is brought to the emergency department with 2 days of fever and painful lumps in his legs bilaterally. In addition, he says that his arms and legs are extremely itchy. Since birth he has had recurrent skin and soft tissue infections. Physical exam reveals a pruritic erythematous scaling rash along both upper and lower extremities bilaterally. Palpation of the painful lesions reveal indurated tissue without any production of pus. Which of the following protein functions is most likely disrupted in this patient?Answer Choices:A. B-cell survival during selectionB. Phagolysosome formation and developmentC. Reactive oxygen species productionD. Transcription factor activity

Reasoning + Answer:

<think> The symptoms of the patient suggest ataxia telangiectasia which is a disorder that affects the signaling of the p53 protein which is a transcription factor and causes the cells to be more prone to cancer and have a decreased survival and the p53 is a transcription factor that is not working properly in the patient with the symptoms of ataxia telangiectasia and defects in the dna repair and cell cycle regulation </think> <answer> D. Transcription factor activity </answer>